R16

Code No: 136EV

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, May - 2019
DATA STRUCTURES
(Common to EEE, ECE, CSE)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

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PART - A

(25 Marks)

1.a) Classify the types of data structures. [2]
b) What is an algorithm? What criteria it has to satisfy. [3]
c) Define a stack. What operations can be performed on it? [2]
d) By using the stack find the value for the postfix expression 242/*68*- [3]
e) How will you find the maximum no.of nodes present at a particular level of a binary tree? [2]
f) Distinguish between complete and full binary tree. [3]
g) Write the average, worst and best case time complexity for linear search. [2]
h) What is meant by collision? How will you resolve it? [3]
i) What is a skewed tree? Give example [2]
j) Write the properties of a typical Red – Black Tree. [3]

PART - B

(50 Marks)

2. With the help of suitable example explain the various asymptotic notations used in algorithms. [10]

OR

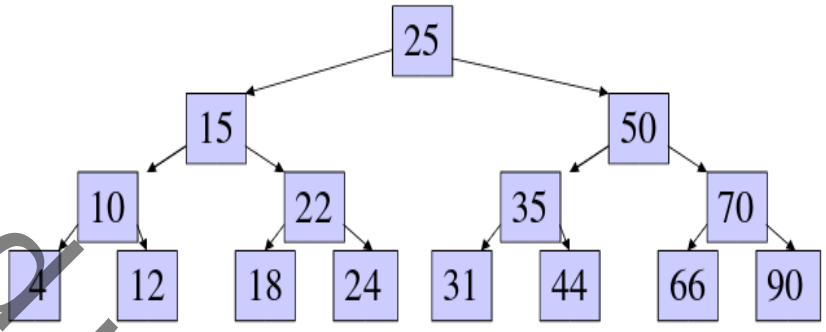
3. With the help of a neat sketch and example explain the various possible positions to insert a node into a double linked list. [10]

4. Write a procedure to convert any given infix expression into postfix form? Using step by step procedure, convert the following infix expression into postfix form: [10]
A + (B * C)-((D*E+F)/G)

OR

5.a) Write and explain the different types of double ended queues.
b) Give brief description about the queue storage using arrays. [5+5]

6. What is meant by tree traversal? List the different traversal used in trees? Find the different orders for the following binary tree. [10]



OR

7. What is a graph? Where you are going to use the graphs? With the help of an example explain the different techniques to represent a graph. [10]

8. Write the procedure to sort the set of elements by using Quick sort? Apply quick sort mechanism to sort the following data {J, N, T, U, H, E, X, A, M}. [10]

OR

9. Apply max heap sort on the list of elements 14, 12, 9, 8, 7, 10, 18, 20, 30. [10]

10. What are the limitations of a typical binary search tree? Explain L – L and R – R rotations with an example. [10]

OR

11. Explain the KMP pattern matching algorithm with example. [10]

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